

Project Starline: A high-fidelity telepresence system

Jason Lawrence
Google
Seattle, United States
jdlaw@google.com

Ryan Overbeck
Google
San Francisco, United States
rover@google.com

Todd Prives
Google
San Francisco, United States
prives@google.com

Tommy Fortes
Google
Seattle, United States
tfortes@google.com

Nikki Roth
Google
Mountain View, United States
nikkiroth@google.com

Brett Newman
Google
New York, United States
brettnewman@google.com

ABSTRACT

Experience Project Starline, the first photorealistic telepresence system that demonstrably outperforms 2D videoconferencing systems, as measured by participant ratings, meeting recall, and non-verbal behaviors [Lawrence et al. 2021]. Our prototype communication endstations combine state-of-the-art face tracking, real-time neural view synthesis, spatial audio, multi-stream compression, and a high-resolution auto-stereoscopic display to produce a striking and natural sense of *co-presence* between two meeting participants that is not possible with traditional videoconferencing.

KEYWORDS

telepresence, real-time, AR/VR/MR

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1 INTRODUCTION

For over 100 years, the telecommunications field has steadily increased both the fidelity and availability of synchronous communication over long distance networks. This includes the advent of video-based communication systems that have led to products like Meet, Zoom, and Teams; the development of purpose-built *immersive teleconferencing* or *telepresence* systems equipped with carefully designed furniture, displays, cameras, microphones, and speakers; and, more recently, systems that employ virtual- or augmented-reality head-mounted displays. However, today's state-of-the-art communication technology still does not reproduce many of the benefits of meeting in-person.

We have developed a new real-time symmetric bidirectional system designed for 1:1 meetings that achieves a much stronger sensation of *co-presence* than prior 2D and 3D video communication



Figure 1: Two people connecting through Project Starline prototypes. Each person sees the other at a very high audiovisual fidelity, at natural scale, and with the full range of non-verbal cues that are so important in natural human conversation.

systems. We carefully designed the physical layout, lighting, 3D imaging, display, and rendering, compression, and audio subsystems to convince the user's senses that the remote participant is sharing their same physical space. As a result, users of our prototype systems report experiencing a strong sense of presence and connection with the remote participant. We measured signals known to indicate positive meeting dynamics, including turn-taking, body language, and eye contact and found higher rates than in a baseline 2D system [Lawrence et al. 2021]. Our user research also shows that people using our system experience a higher recall of meeting contents than in a traditional 2D videoconferencing system.

Project Starline was announced publicly in May of 2021,¹ and the overall system design was described in Lawrence et al. [2021]. In 2023, we revealed a new prototype version with a simpler mechanical design, and fewer and simpler hardware sensors. We are excited to offer demos of this new prototype at SIGGRAPH 2024's Emerging Technologies exhibit.

2 PRIOR WORK

Through rigorous user studies, we have learned that multiple factors are important for creating an effective telepresence experience. This includes 3D depth cues like stereopsis and motion parallax, but equally important are natural scale, eye contact, symmetric bidirectional communication, high resolution photorealistic imaging, high framerates (60Hz) with low latency, and clear spatial audio.

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¹<https://blog.google/technology/research/project-starline/>

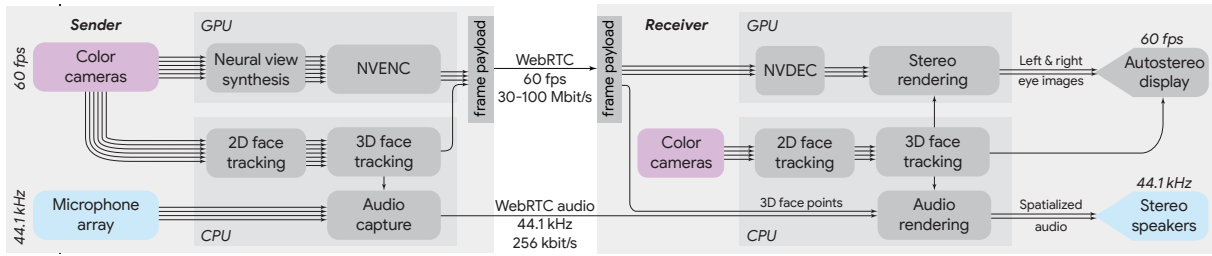


Figure 2: Overview of the main processing components and data flow in our system.

A number of commercial products have attempted to improve the 2D videoconferencing experience with custom furniture and specially designed configurations of displays, cameras, microphones, and speakers [Hewlett-Packard 2007; Szigeti et al. 2009]. Other works demonstrated new ways of presenting a 3D image without requiring special glasses or a headset [Jones et al. 2009, 2015; Nagano et al. 2013]. Additionally, there have been many demonstrations of real-time 3D reconstruction that show steady improvements in quality over time [Maimone et al. 2012; Stengel et al. 2023]. Altogether, none of these prior works addressed all aspects required for effective telepresence, nor did they provide measurements or studies to prove a clear benefit over existing 2D systems.

3 SYSTEM OVERVIEW

Much of our overall system design and user studies were first described by Lawrence et al. [2021]. The prototype that we will demonstrate at SIGGRAPH 2024 was first shared in 2023.² This prototype has a more streamlined design and significantly fewer cameras. Specifically, the 3d imaging subsystem is now powered by six regular RGB video cameras and uses a ML-based technique to perform real-time view synthesis. This neural view synthesis capability replaces the active depth-from-stereo approach and associated infrared cameras and infrared light sources in our earlier prototype.

Figure 2 describes the major processing components and data flow in our system. Our system combines a head-tracked autostereo display with a novel 3D eye and ear tracking subsystem that combines 2D facial landmark estimation, 3D triangulation, and double-exponential filtering to provide accurate predictions at 60 Hz. It also includes an audio subsystem that combines a beamforming microphone array with stereo loudspeakers to provide a free-space spatialized experience guided by real-time 3D tracking. Finally, our 3d imaging, compression, and rendering systems achieve an unprecedented level of visual fidelity at real-time video rates (60Hz) over standard networks.

4 DEMO EXPERIENCE

Our demo consists of two identical prototype endstations, each located within a surrounding "demo pod." Each pod provides a light- and sound-controlled space in which to experience the demo. Attendees will experience our demo one-at-a-time, whereby they will enter one of these pods and converse with a member of our team who is located in another pod. During the demo, they will

experience the basic functionality of our system (e.g. real-time high-fidelity audio and 3D video of their meeting partner with mutual eye gaze, non-verbal cues, etc.). We will also share two "debug" display modes that showcase our real-time facial interest point tracking system and a false-color visualization of the displayed 3d geometry, respectively. The experience will last roughly 2-3 minutes, and we will invite participants to complete a survey afterwards to contribute to our understanding of the benefits and limitations of our technology. In addition, our exhibit will feature a large TV screen describing our project and prior research, so that attendees may learn about our project while waiting for their demo.

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²<https://blog.google/technology/research/project-starline-prototype/>